

QUESTION:11QUEUE USING ARRAY

main.c

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```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #define MAX 5
4 int queue[MAX];
5 int front = -1;
6 int rear = -1;
7 void enqueue(int value) {
8     if (rear == MAX - 1) {
9         printf("Queue Overflow\n");
10    } else {
11        if (front == -1) {
12            front = 0;
13        }
14        rear++;
15        queue[rear] = value;
16        printf("%d inserted into queue\n", value);
17    }
18 }
19 void dequeue() {
20     if (front == -1 || front > rear) {
21         printf("Queue Underflow\n");
22     } else {
23         printf("Deleted element: %d\n", queue[front]);
24         front++;
25         if (front > rear) {
26             front = rear = -1;
27         }
28     }
29 }
```

Output

```
--- QUEUE USING ARRAY ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value: 3
3 inserted into queue

--- QUEUE USING ARRAY ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted element: 3

--- QUEUE USING ARRAY ---
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: |
```

main.c



Run

```
27     }
28 }
29 }
30 void display() {
31     if (front == -1 || front > rear) {
32         printf("Queue is Empty\n");
33     } else {
34         printf("Queue elements are:\n");
35         for (int i = front; i <= rear; i++) {
36             printf("%d ", queue[i]);
37         }
38         printf("\n");
39     }
40 }
41 int main() {
42     int choice, value;
43     while (1) {
44         printf("\n--- QUEUE USING ARRAY ---\n");
45         printf("1. Enqueue\n");
46         printf("2. Dequeue\n");
47         printf("3. Display\n");
48         printf("4. Exit\n");
49         printf("Enter your choice: ");
50         scanf("%d", &choice);
51         switch (choice) {
52             case 1:
```

```
53         printf("Enter value: ");
54         scanf("%d", &value);
55         enqueue(value);
56         break;
57     case 2:
58         dequeue();
59         break;
60     case 3:
61         display();
62         break;
63     case 4:
64         exit(0);
65     default:
66         printf("Invalid Choice\n");
67     }
68 }
69 return 0;
70 }
```